TAHA ASIF

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PROFILE

B.Sc. student in Infotronic Systems Engineering and Student Research Engineer at the Centre for Assistive Technology Rhine-Ruhr (ZAT), Rhine-Waal University of Applied Sciences. Developing assistive AI systems: RAG pipelines, local LLM infrastructure (Ollama), and adaptive desktop applications with Python/Qt. Co-author of an accepted paper at the "Mensch und Computer 2026" (MuC) conference.

WORK EXPERIENCE

Student Research Engineer (HiWi) – Centre for Assistive Technology Rhine-Ruhr (ZAT) 03/2026 – Present

- Developing a pro-adaptive reading support system for individuals with ADHD: adaptive control logic (Dynamic Support Index), eye-tracking integration (Tobii), and LSL data streaming
- Architecting local LLM gateways and vector database pipelines (Ollama, ChromaDB) for privacy-compliant on-device processing
- Engineering PDF/EPUB parsing pipelines (Docling, PyMuPDF4LLM) for structured document ingestion into LLMs

Student Assistant & AStA Event Lead – Rhine-Waal University of Applied Sciences 03/2024 – Present

- Planning and executing large-scale student events including budget management; primary liaison between the student body and university administration

PROJECTS

ADHD Reading Assistant – ZAT / Rhine-Waal University of Applied Sciences Since 03/2026

Python, PySide6/Qt, Tobii Eye-Tracking, LSL, Docling

- Desktop reading application (PySide6/Qt) with eye-tracking integration and Lab Streaming Layer streaming
- Implemented adaptive "Dynamic Support Index": reading speed and medication trajectory govern five adaptive visual support tiers (EMA smoothing, hysteresis threshold)
- Document parsing pipelines for PDF, EPUB, and web content for LLM-assisted text simplification

HEXILEXI – AI Word Detective – Project Lead & Lead Developer 2025 – 2026

React, TypeScript, Ollama (Gemma 3:12b), Ant Design

- LLM-driven active vocabulary mentor for primary school children, integrated as a React component into the public STREEN platform (streen-app.ais-lab.de)
- Two-step prompt pipeline ("Architect/Builder") via local Ollama instances; frontend fuzzy-matching evaluation (Levenshtein distance) replacing resource-heavy LLM grading
- Evaluated across four test iterations under real classroom conditions in two primary schools; UI state-locking and request queues for stable concurrent usage

Movella DOT → LSL Pipeline (Open Source) – Maintainer Since 2026

Python, BLE, Lab Streaming Layer, uv

- Extended and maintained real-time IMU streaming pipeline (16 channels, 30 Hz) for the ZAT Rhine-Ruhr project (github.com/taaha17/movella-dot-lsl-pipeline)
- Example viewers, diagnostic tools, and setup documentation for Ubuntu 22.04 and Windows

PUBLICATIONS

H. Ashfaque, T. Asif, A. F. Krause, N. Wild-Wall, C. Ressel, K. Kannen: "ADHD by Design: Expert Co-Design and Prototype Development Towards a Pro-Adaptive Reading Support System". Workshop "Towards Pro-Adaptivity in Human-Computer-Interaction", Mensch und Computer 2026, Duisburg (Aug 30, 2026). In Press (GI Digital Library).

EDUCATION

B.Sc. Infotronic Systems Engineering – Rhine-Waal University of Applied Sciences Since 10/2023 · Expected 2027

A-Levels, Computer Science – Lahore Grammar School, Lahore (Pakistan) 2021 – 2023

TECHNICAL SKILLS

Programming: Python, TypeScript, Kotlin, Java
AI / LLM: Ollama, LangChain, ChromaDB, RAG, Prompt Engineering, Agentic Workflows
Web & Databases: React, Next.js, PostgreSQL, MongoDB
Tools & Infrastructure: Git, Docker, Linux (Ubuntu), uv
Desktop & Hardware: PySide6/Qt, Pygame, Eye-Tracking (Tobii), Lab Streaming Layer

LANGUAGES

English (C2 / Proficient) · German (A2 / Elementary)